

1. This part of sheet music appears right after the key signature. The top number tells how many beats are in a measure, while the bottom number means what kind of note is worth one beat. What is this musical item that looks like a fraction?

TIME SIGNATURE

2. This graphic novel's protagonist, Jordan Banks, loves to draw cartoons. He encounters racism when his parents enroll him at the Riverdale Academy Day School. Identify this Newbery winner by Jerry Craft.

NEW KID

3. These numbers are types of real numbers. We use these numbers to count. What kind of numbers beginning with the letter "N" start with 1?

NATURAL (NUMBERS)

4. Two words, please. For water at sea level, this temperature is 212 degrees Fahrenheit. What is this term for the temperature at which a substance rapidly evaporates?

BOILING POINT

5. The Pillars of Hercules and Frozen Niagara are among the names given to this national park's many stalactites and stalagmites. Name this significant cave system in west-central Kentucky, the state's only national park.

MAMMOTH CAVE (NATIONAL PARK)

6. This positive number is a factor of every real number. What positive number is a factor of both 29 and 2?

1 (ONE)

7. This wordless picture book's heroine is a white dog with a brown tail and ears. When a bigger dog accidentally destroys her favorite toy, the dog's owner buys her a replacement. Give the title of this second Caldecott-winning book illustrated by Chris Raschka [ROSH-kuh].

(A) BALL FOR DAISY

8. He led Kentucky's militia during the Battle of the Thames, an engagement of the War of 1812, which occurred while he served as Kentucky's fifth governor. Name this political leader who also served as Kentucky's first governor.

(ISAAC) SHELBY

9. One of these string ensembles consists of two violins, a viola, and a cello. For voices, barbershop ones perform a cappella, meaning without musical accompaniment. Give the word for a piece written for, or a group consisting of, four parts.

QUARTET

10. This type of energy depends on the sum of an object's kinetic and potential energy. When steam, water, wind, or gas produces it, it can be converted into power. What is this energy of motion?

MECHANICAL (ENERGY)

11. This heat transfer method occurs when part of the electromagnetic spectrum is emitted or absorbed. As with the sun, it often involves infrared and ultraviolet rays. Identify this counterpart of conduction and convection.

RADIATION

12. This piano style was sometimes called the "fast Western-style" or "Western rolling blues." Like ragtime before it, it was popular with honky-tonks, but its popularity declined after World War Two. Give the rhyming name for this music genre.

BOOGIE-WOOGIE

13. One example of this type of landform in the United States is Delmarva, where most of the state of Delaware sits. Other examples include upper and lower Michigan as well as Florida. Give the term for a piece of land that is surrounded by water on three of its sides.

PENINSULA

14. This poet used alliteration and assonance in "The Bells." The death of his wife, Virginia Clemm, likely inspired his poem "Annabel Lee." Name this poet of "The Raven."

(EDGAR ALLAN) POE

15. A farmer has 2 and 1/2 acres planted in corn and 3 and 1/4 acres planted in wheat. How many total acres does the farmer have planted in corn and wheat?

5 3/4 (FIVE AND THREE-FOURTHS ACRES) (ACCEPT: 5.75 (ACRES))

16. This instrument's name is German for "set of bells." It has metal bars arranged in two rows that are struck with mallets or hammers. Name this high-pitched percussion instrument that looks like a small xylophone.

GLOCKENSPIEL [GLOCK-in-shpeel]

17. At age 6, this novel's title character has read classics like *Oliver Twist* and *Pride and Prejudice*. Her kind teacher, Miss Honey, adopts her after her parents move to Spain. Name this Roald Dahl [ROO-all DOLL] novel about a young girl with telekinetic powers.

MATILDA

18. On this scale, glass rates a 5.5, while a fingernail rates a 2.5. Developed in 1812, it measures a smooth surface's resistance to abrasion or scratching. Identify this scale for measuring a mineral's hardness.

MOHS (HARDNESS SCALE)

19. Examples of this two-word alliterative term include hemp, cotton, and tobacco. What term refers to any agricultural product grown to sell for a profit?

CASH CROP

20. Helen baked 24 cookies. She cut 18 of the cookies into heart shapes and 6 into circles. What is the ratio of heart-shaped cookies to the total number of cookies?

3:4 (THREE TO FOUR)

21. Like science fiction, this literary genre depends on strange settings and characters. At least some of its events, places, and creatures could never occur in real life. Name this imaginative genre of *The Chronicles of Narnia* and *The Hobbit*.

FANTASY

22. While held in captivity in 1613, she was converted to Christianity and took the name, Rebecca. Identify the daughter of the powerful chief Powhattan whose conversion, marriage to John Rolfe, and travels to England made her a celebrity in her time.

POCAHONTAS

23. This father of 20 children wrote a passion for each of the Four Gospels. His *Goldberg Variations* may have been named for their first performer. Name the Baroque composer of the *Brandenburg Concertos*.

(JOHANN SEBASTIAN) BACH

24. These high-speed winds form at the boundary of cold and hot air masses. The primary examples are polar and subtropical and occur between 6 and 9 miles above the Earth. What are these narrow, shallow winds named for the airplanes they can help propel?

JET STREAM(S)

25. A football field is 100 yards long. How many feet long is the field?

300 (THREE HUNDRED FEET)

26. A triangle has three sides and three angles. One of the angles is larger than 90 degrees. Based on the size of the angles, what type of triangle is being described?

OBTUSE (TRIANGLE)

27. In this fable, a fox agrees to judge a contest between the title characters. One of the competitors naps while the other crawls slowly and steadily toward the finish line. Give the title of this Aesop fable illustrating, "The race does not always go to the swift."

(THE) TORTOISE AND THE HARE (Accept (THE) HARE AND THE TORTOISE)

28. Actions taken in April 1775 by this silversmith were immortalized in a famous poem by Henry Wadsworth Longfellow. Name the American Founding Father remembered for his "midnight ride," in which he alerted Boston residents and militia members that the British were coming.

(PAUL) REVERE

29. This costume extended to the dancer's calf in the early Romantic Era. The classical type was shortened to above the knee, while the powder puff kind lacked a hoop. Give the name of this full ballet skirt made of tulle.

TUTU

30. This type of astronomical object consists of a nucleus and, when it nears the sun, a coma, and a tail. It is a chunk of ice and dust that some people call a dirty snowball. What is this kind of body, whose most famous example is named for Edmond Halley?

COMET(S)

End of First Half – Begin Three-Minute Inquiry Period

Round 1 - Second Half

Elementary – 2023 District



31. Cells make seven types of these polymers, including enzymes and antibodies. They form from long chains of amino acids. What is this organic molecule that also helps build muscle?

PROTEIN

32. This woman choreographed the musicals *Paint Your Wagon*, *Brigadoon*, and *Carousel*. She also created dances for the ballets *Fall River Legend* and *Rodeo*. Name this choreographer of Rodgers and Hammerstein's *Oklahoma!*

(AGNES) DE MILLE

33. As she does in the Disney movie, this fairy tale character eats a poisoned apple and is placed in a glass coffin. A prince's kiss awakens her, and the evil queen who poisoned her dies at this character's wedding. Which Brothers Grimm creation befriends the Seven Dwarves?

SNOW WHITE

34. The Battle of Queenston Heights was this War's first significant engagement, resulting in a British victory. Name this War fought between Great Britain and the United States, which featured the burning of the U.S. capital.

WAR OF 1812

35. Jeffrey is planning to build a safety fence around his pool. His rectangular pool measures 10 feet by 16 feet. How many feet of fencing will he need to fence the entire pool?

52 (FIFTY-TWO FEET)

36. Shakespearean plays in this genre include *The Taming of the Shrew* and *A Midsummer Night's Dream*. Slapstick, parody, and farce are all types. Give the word for a funny play with a happy ending.

COMEDY

37. Correct the following sentence, if necessary: She ran the race really good.

CHANGE "GOOD" TO "WELL" (Accept equivalents)

38. Harrison is trying to choose what to wear to school today with his red sweatshirt. He has 5 pairs of long pants and 3 pairs of shorts that will match the sweatshirt. How many total choices does he have?

8 (CHOICES)

39. All major cereal grains, legumes, and many gourds fit into this plant category. It is the counterpart of biennial and perennial plants. Give the term for a plant that completes its life cycle in a single growing season.

ANNUAL (PLANT)

40. Of the United States' plan to go to this location, John F. Kennedy said, "We do these things not because they are easy but because they are hard." Name this location reached by the United States for the first time on July 20, 1969.

(THE) MOON

41. One word pronounced this way is a plural possessive pronoun meaning "belonging to us." Another word with this pronunciation is a unit of time made up of 60 minutes. What are these homophones?

OUR/HOUR (ACCEPT EITHER ONE OR BOTH)

42. As of January 2023, Kentucky is represented in this government body by Rand Paul and Mitch McConnell. Name this upper chamber of Congress where each state elects two members to represent them.

(UNITED STATES) SENATE

43. Use an addition sign in your answer. Sally spent two days of her holiday vacation visiting her aunt and three days visiting her grandmother. What numerical expression would show how many days she spent visiting her relatives?

$2 + 3$ (ACCEPT: $3 + 2$)

44. In certain species, special names for this metamorphic stage include grub, maggot, nymph, and caterpillar. In frogs, tadpoles are at this stage. Give the name for this step that, for most insects, occurs between egg and pupa.

LARVA(L) (STAGE)

45. In encaustic painting, this substance has colored pigments added and is then heated. It forms the cores of some colored pencils and most crayons. Identify this substance made from petroleum, plant, or animal sources, including bees.

WAX

46. Use x as the variable to stand for the total number of minutes Kelly spends at the mall. She spends 30 minutes in each store, and she visits 6 stores. What algebraic equation could be used to solve the problem?

$30 \text{ TIMES } 6 \text{ EQUALS } X$ (ACCEPT: $6 \text{ TIMES } 30 \text{ EQUALS } X$ or $X = 30 \text{ TIMES SIX}$)

47. This art form resembles photomontage [foh-toh-mahn-TAHZH] and assemblage [ah-sahn-BLAHZH]. Georges Braque [ZHAWRZH BROCK] and Pablo Picasso invented it during their Synthetic Cubist phase. Name this collection of materials like newspaper, fabric, and string glued onto paper to make a picture.

COLLAGE

48. Among the animals who engage in this activity are marmots, ground squirrels, hedgehogs, and bats. Amphibians, reptiles, insects, and bears may behave similarly, but their body temperatures and breathing rates remain almost constant. What is this period of dormancy that some animals experience during the winter months?

HIBERNATION (Accept word forms)

49. Millions of Americans demonstrate this right yearly on the first Tuesday in November. Give the name for this right to participate in public elections, a right that Susan B. Anthony worked to gain on behalf of women.

(RIGHT TO) VOTE (ACCEPT: VOTING or SUFFRAGE or (POLITICAL) FRANCHISE)

50. This punctuation mark, one of two possible ends for an imperative sentence, is called a "full stop" in Britain. Name this punctuation that always concludes a declarative sentence.

PERIOD

51. What property of real numbers is illustrated by the following equation? Six plus eight equals eight plus six

COMMUTATIVE (PROPERTY OF ADDITION)

52. This two-word architectural feature was common in Gothic cathedrals. It consists of a supportive arch that sticks out from an exterior wall. What is this feature that, despite its name, doesn't soar through the air?

FLYING BUTTRESS(ES)

53. Exploring this science often involves studying such factors as production, resources, and the creation and consumption of goods and services. Name this science, usually broken down into two main branches of study: micro and macro.

ECONOMICS

54. The largest one of these blood vessels is the aorta. Carotid ones reside in the head and neck. Identify these large vessels that carry blood away from the heart.

ARTERY (Accept word forms)

55. This part of speech is collective if it refers to a group. All examples are either proper or common, depending on whether they name specific or general entities. Identify the part of speech that names a person, place, thing, or idea.

NOUN(S)

56. Many of these sentences begin with one of journalism's five Ws. Those words are "who," "what," "when," "where," and "why." What kind of sentence asks a question?

INTERROGATIVE (SENTENCE)

57. In today's laboratories, this equipment has racks to hold culture dishes or flasks. Different models may control light, air circulation, humidity, oxygen, and temperature. What is this enclosed chamber that can help protect eggs or human babies?

INCUBATOR

58. There are 10 pencils in the box. You take out 3 pencils. What percent of the pencils did you remove from the box?

30% (THIRTY PERCENT)

59. This list was initially comprised to resolve several issues that Anti-Federalists had with the U.S. Constitution. Included within the list are constitutional amendments which address such issues as the freedom of speech and the right to keep and bear arms. Give the name for the first ten amendments to the U.S. Constitution.

BILL OF RIGHTS

60. This artist was a war correspondent during the Spanish-American War. Many of his works depict Native Americans, soldiers, horses, and other aspects of Western U.S. life in the 1800s. Name this artist remembered for bronze sculptures like *The Bronco Buster*.

(FREDERIC) REMINGTON

End of Second Half – Begin Three-Minute Inquiry Period

Extra Questions

61. Traditionally, this boat was a one-person hunting craft. It had a wooden frame, a closed deck, and an exterior covered by caribou or seal skin. Give the name of this Inuit boat propelled with a single- or double-bladed paddle.

KAYAK

62. Spelling required. I can't stomach my sister's whining and complaining. Spell the body part in the previous sentence.

S-T-O-M-A-C-H

63. There are three yellow balls and eight red balls in a bag. What is the probability that you will pull a red ball from the bag on the first try?

8/11 (EIGHT-ELEVENTHS)

64. This man developed vaccines against rabies and anthrax. He also discovered that microbes caused fermentation and devised a way to kill those microbes. Name this man who heated food to slow its spoilage rate.

(LOUIS) PASTEUR [pas-TUR]

65. Following the election of President Joe Biden, this American physician was asked to become the president's chief medical advisor, a position he would fill while remaining director of the National Institute of Allergy and Infectious Diseases. Name the doctor who became internationally known as the White House Coronavirus Task Force leader.

(DOCTOR) (ANTHONY "TONY") FAUCI [FOU-chee]